Human-AI Interaction for Augmented Reasoning: Improving Human Reflective and Critical Thinking with Artificial Intelligence

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Figure 1: Overview of potential applications of AI-enhanced reasoning systems

Abstract

AI-Augmented Reasoning systems are cognitive assistants that support human reasoning by providing AI-based feedback that can help users improve their critical reasoning skills. Made possible with new techniques like argumentation mining, fact-checking, crowdsourcing, attention nudging, and large language models, AI augmented reasoning systems can provide real-time feedback on logical reasoning, help users identify and avoid flawed arguments and misinformation, suggest counter-arguments, provide evidence-based explanations, and foster deeper reflection. The goal of this workshop is to bring together researchers from AI, HCI, cognitive and social science to discuss recent advances in AI-augmented reasoning, to identify open problems in this area, and to cultivate an emerging community on this important topic.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interactive systems and tools**; **HCI design and evaluation methods**.

Keywords

Human-AI Interaction, Augmented Reasoning, Intelligence Augmentation, Cognitive Enhancement

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1 Motivation

We are increasingly exposed to massive amounts of information that can be potentially deceptive, misleading, or strictly false. To process this information, researchers have shown that people frequently use "cognitive shortcuts" that do not involve critical, logical and cautious thinking as people are often strongly affected by social influences, group affiliations, charismatic leaders, personality traits, and rhetorical manipulation [6, 9, 11, 13, 21, 23, 24]. In such cases, critical thinking becomes an important skill to accurately process the information we encounter and integrate it into our existing beliefs and behaviors. For example, research has shown that reflective reasoning skills increase human abilities to detect misleading information as it allow us to override otherwise unfounded intuitions [5, 12, 16, 17, 19, 22]. Despite the importance of reflective reasoning, it is a skill not everyone masters nor have the cognitive resources and time to learn and apply [6, 11, 13, 21, 23, 24].

In this context, there is a pressing need for tools and strategies to help individuals navigate the torrent of information and develop critical thinking skills that can empower them to discern fact from fiction. In order to help people think more critically, the emerging field of AI-enhanced reasoning has proposed the development of AI interventions that promote critical thinking based on the study of human reasoning [2–4, 18, 20]. In this context, researchers have proposed that AI-based reasoning should be based on logic analysis, fact-checking, attention nudging, and the correct type of human-AI interaction (see Fig 2). For example, some researchers have developed cognitive assistants that teach users about logical fallacies [14, 15], while others have developed systems that give real-time feedback on logical structures of arguments [2]. Fact-checking systems have also been developed to detect claims that are false or misleading [7, 8], attention nudging systems have been developed to help users in identifying misleading information [20], and human-AI interaction methods such as Socratic questioning to make sure that the user is motivated to engage critically [3, 10]. Despite these advances, however, there are still many open questions that need to be addressed to develop more effective AI-based reasoning interventions.

(1) How do people think critically and what are the most critical reasoning skills that need to be developed and promoted?
(2) How can AI-based reasoning interventions be designed to effectively promote critical reasoning skills?
(3) How may AI-based reasoning interventions lead to better critical thinking in the long-term?
(4) What methods for explainability and interaction between a human and AI system are most effective in inducing critical thinking?
(5) How can AI systems promote internalization (learning) of critical reasoning skills, rather than creating system dependency (and should they)?
(6) How do we evaluate critical thinking? How do we know that a system improves critical thinking.

The goal of this workshop is to bring together researchers in cognitive and social science, AI, and Human-Computer Interaction (HCI) to discuss recent advances in AI-based reasoning and to identify open problems in this area.

1.1 Definition

AI-enhanced reasoning refers to the application of artificial intelligence (AI) systems to support, facilitate and improve human reasoning capabilities by providing insights, identifying patterns, uncovering biases, and offering guidance that is intuitive to the user and which enables them to make better-informed decisions that they feel that they arrived at through their own thinking processes.

AI-enhanced reasoning systems differ from other AI information-processing systems in that they focus not only on providing accurate information or optimize decision outcomes but also on actively engaging users in reflective thinking. In order to be considered an AI-enhanced reasoning system, the following conditions should be met:

(1) Interaction: The AI system should *interact* with users in a manner that fosters critical thinking and reflection when appropriate, like using Socratic questioning, argument graphs, the correct timing, divergent perspectives, etc.
(2) Analysis: The system should provide guidance and support based on accurate evaluations of information, such as identification of logical fallacies, detection of biases, or checking of facts, or potential issues.
(3) Personalization: The AI system should be adaptive and responsive to the user's needs, offering personalized feedback and suggestions based on the user's cognitive abilities and reasoning styles.

1.2 Applications

With the adaptation and usage of AI across industries (See Fig. 1), we see a potential for future positive applications within a variety of fields such as:

(1) Political Reasoning: AI-enhanced reasoning systems can support civic engagement by helping citizens critically evaluate political information, understand different perspectives, and make informed decisions. These systems can provide evidence-based analysis of policy proposals, identify logical flaws in political rhetoric, and encourage constructive dialogue among citizens.
(2) Misinformation Online: AI-augmented reasoning can be integrated into social media platforms to enhance users' critical thinking skills. These systems can provide real-time feedback on the veracity of shared content, highlight potential biases, and encourage users to reflect on the information before sharing it further, while over time building information literacy.
(3) Health Care: In healthcare, AI systems can assist doctors and medical professionals in diagnosing complex cases by

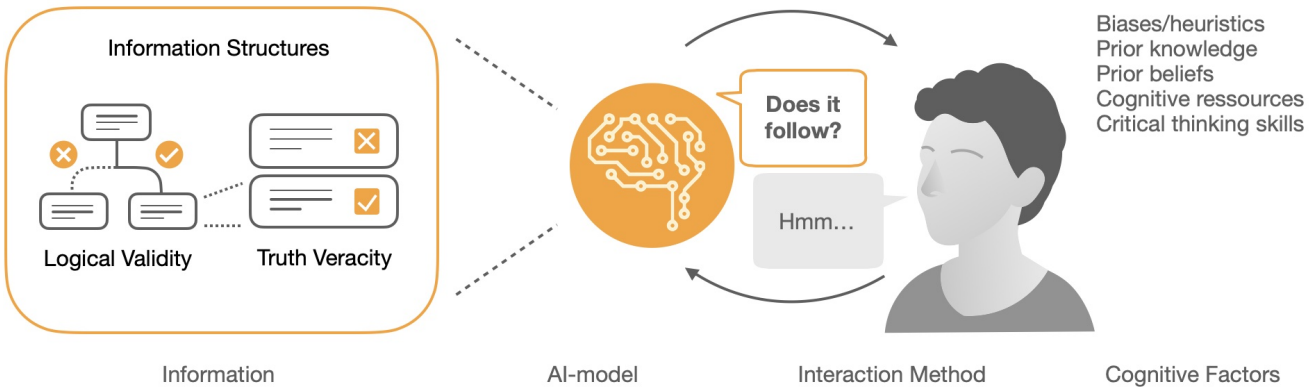


Figure 2: Factors of AI-Enhanced Reasoning

helping them analyze patient data and medical literature instantaneously. These systems can question the evidence in recommendations and suggest alternative perspectives, identify potential biases in decision-making and AI-collaboration, and encourage reflective reasoning, thereby improving patient outcomes.

(4) Education: Critical thinking is an essential skill that is often challenging to incorporate into classroom curriculum due to a lack of resources. AI-Augmented Reasoning systems can support instructors in creating thought-provoking assignments and act as interactive tutoring support for students to better develop problem-solving and critical thinking skills.
(5) Personal Decision-Making: Individuals can use AI-augmented reasoning systems to make well-informed decisions in their daily lives. Whether it's planning finances, making health-related choices, making informed purchasing decisions, negotiating better deals, avoiding scams, or navigating social issues, these systems could offer personalized advice, highlight potential pitfalls, and encourage reflective thinking.
(6) Scientific Research: Researchers can use AI-augmented reasoning tools to analyze vast amounts of scientific data, identify patterns, and generate novel hypotheses. These systems can provide critical feedback on research methodologies, highlight potential biases, and encourage reflective thinking, thereby advancing scientific knowledge.

We foresee a future where AI-augmented reasoning systems will become ubiquitous in our daily lives as part of the ever growing human-AI interaction landscape — but not without challenges. For example, research has shown that humans interacting with AI systems tend to over-rely on the AI system's recommendations and often fail to critical reasoning skills and the ability to think for themselves [1]. As AI systems become more sophisticated and ubiquitous, there is a risk that people will become increasingly reliant on these systems and less capable of thinking for themselves. How do we build reasoning systems that maintain human agency and autonomy while still providing the benefits of AI? What are the ethical and social implications of AI-augmented reasoning systems? How do we ensure that these systems are used for good and not for harm?

In this workshop, we will bring together researchers in HCI, AI and related fields to explore and discuss applications, methods and design considerations for AI-augmented reasoning systems. The goals of the workshop are as follows:

(1) Share State-of-the-Art Research: Present and discuss recent advances in AI-augmented reasoning and human-AI interaction. This includes new techniques, tools, and methodologies developed to enhance critical and reflective thinking.
(2) Identify Challenges and Opportunities: Highlight the current challenges, limitations, and potential risks associated with AI-augmented reasoning systems. Discuss opportunities for future research and development.
(3) Interdisciplinary Collaboration: Foster collaboration between researchers from AI, HCI, cognitive science, social science, and other relevant fields to create a multidisciplinary approach to developing AI-augmented reasoning systems.
(4) Ethical and Social Implications: Delve into the ethical and social issues surrounding AI-augmented reasoning like accidental overreliance, and potential misuses of modeling user reasoning.
(5) Design Principles and Guidelines: Develop design principles and guidelines for creating AI-augmented reasoning systems that prioritize human agency, autonomy, and long-term learning. Consider the balance between AI assistance and human decision-making.
(6) Evaluation Methods: Discuss and propose effective methods for evaluating the effectiveness of AI-augmented reasoning systems. This includes metrics for assessing improvement in critical thinking skills and the impact on decision-making quality.

2 Organizers

2.1 Valdemar Danry

Valdemar Danry is a PhD researcher in the MIT Media Lab Fluid Interfaces research group led by Professor Pattie Maes at Massachusetts Institute of Technology. His research lies at the intersection of philosophy, technology, and the arts. With his background in cognitive science, formal logic, philosophy, and machine learning, he seeks to integrate cognition, especially reasoning, with artificial

intelligence. Valdemar has worked with international collaborators from MIT, Harvard, Georgia Tech, IBM Research, and University of California.

2.2 Pat Pataranutaporn

Pat Pataranutaporn is a multi-disciplinary technologist at the Massachusetts Institute of Technology (MIT). He is a Ph.D. student in the Media Lab's Fluid Interfaces research group. Pat's research lies at the intersection of artificial intelligence, human-computer interaction and wearable computing, specifically the area of human-AI symbiosis and cognitive enhancement. Pat has previously worked with global collaborators including NASA TRISH, IBM Research, Bose, ASU, NTU on projects that examine the relationship between people and technology across applications and systems.

2.3 Lancelot Blanchard

Lancelot Blanchard is an AI researcher, engineer and musician from the Department of Computing of Imperial College London. His research interests center around the use of Human-Centered Artificial Intelligence (HCAI) for the enhancement of human reasoning and creativity. He is particularly interested in ways to use Artificial Intelligence as a tool in the generation of wisdom and art, rather than relying on this technology as a unique source of creation. As such, he is interested in the development of interfaces and languages that can help fluidify and strengthen the link between Human users and AI systems.

2.4 Christopher Cui

Christopher Cui is a PhD student in the PEARLS group, led by Prithviraj Ammanabrolu at University of California, San Diego. Previously at the Georgia Intitute of Technology, he worked with professors Mark Riedl in Reinforcement Learning Agents in Textworld Environments, and Thad Starner in Educational Technology tools for large-scale classrooms. His research interests focus on the application of Natural Language Processing and Reinforcement Learning in creating AI agents that work collaboratively with humans to solve tasks.

2.5 Jui-Tse Hung

Jui-Tse Hung is a Master's student and researcher in the School of Interactive Computing at Georgia Institute of Technology. His work focuses on applying Artificial Intelligence and Human-Computer Interaction techniques to develop Educational Technology tools for large-scale classrooms. Ray's research interests include AI-enhanced learning and assessment solutions for improving critical thinking and communication skills.

2.6 Zana Buçinca

Zana Buçinca is a PhD candidate in Computer Science at Harvard working at the intersection of human-AI interaction and responsible AI. Her research integrates cognitive and social science theories to design novel AI-powered decision-support systems that optimize task performance, and decision-maker competence and autonomy by guiding and eliciting meaningful cognitive engagement with AI-generated content. Her research has been recognized with the IBM PhD Fellowship and a Best Paper Award at IUI 2020. Zana was named a Siebel Scholar, a Rising Star in AI and a Rising Star in MS&E.

2.7 Chenhao Tan

Chenhao Tan is an assistant professor of computer science and data science at the University of Chicago, and is also affiliated with the Harris School of Public Policy. He directs the Chicago Human+AI Lab. His goal is to develop the best AI for humans. He has won a Sloan research fellowship, an NSF CAREER award, and research awards from companies such as Google and IBM.

2.8 Thad Starner

Thad Starner is a wearable computing pioneer, having worn a computer with a head-up display in his daily life since 1993. Dr. Starner is a Professor of Computing at Georgia Tech and a Technical Lead on Google Glass. Besides Glass, Thad's projects include a wireless glove that teaches how to play piano melodies without active attention by the wearer; a glove that helps stroke patients recover tone, sensation, and dexterity in their hands; a game for deaf children using computer vision-based sign language recognition that helps them acquire language skills; underwater wearable computers for two-way communication experiments with wild dolphins; wearable computers for working dogs to facilitate communication with their handlers; brain computer interfaces that recover phrase-level sign language from the motor cortex; and systems for recognizing English speech without vocalization.

2.9 Pattie Maes

Pattie Maes is a Professor in MIT's Media Arts and Sciences Program, where until recently she served as Academic Head. Pattie also runs the Media Lab's Fluid Interfaces research group, which aims to radically reinvent the human-machine experience. Coming from a background in artificial intelligence and human-computer interaction, she is particularly interested in the topic of cognitive enhancement or how immersive and wearable systems can actively assist people with memory, attention, learning, decision making, communication, and well-being. Pattie is the editor of three books and serves as an editorial board member and reviewer for numerous professional journals and conferences. Pattie received several awards including an honorary doctorate from the Vrije Universiteit Brussel in Belgium.

3 Plans to Publish Proceedings

We plan to publish a collection of the submitted content after the workshop as workshop proceedings, for example, via https://ceur-ws.org or ArXiv using report numbers.

4 In-person, or hybrid

The workshop will take place in a hybrid format allowing participants to join over video-conferencing platforms. The interactive sessions will be conducted using collaborative platforms such as Miro, allowing both virtual and in-person participant collaboration. For virtual participants, the workshop will be transcribed using the video-conference of choice's built-in transcription tools.

5 Asynchronous Engagement

Participants joining virtually will be able to access asynchronous material such as schedules, shared readings, and notes from the workshop.

6 Workshop Activities

Given the interdisciplinary nature and innovative potential of AI-augmented reasoning, the workshop will emphasize interaction, collaboration, and actionable insights. The condensed one-day structure will focus on key topics and foster enriched dialogue and exploration within a limited time frame.

9:00 AM – 9:15 AM: Introduction and Welcome (Hybrid)

- Overview of workshop goals and structure.
- Introduction to organizers and keynote speaker.
- Brief introduction of participants.

9:15 AM – 10:30 AM: Opening Keynotes (Hybrid)

- Presentation by a leading experts in AI-augmented reasoning.
- Keynote speakers: Dr. Thomas Costello, an expert known for his research on the psychology of reasoning, and AI systems for debunking beliefs in conspiracy theories; and Dr. Paolo Torroni, an expert known for his work in natural language processing on building AI systems that can analyze and evaluate arguments and reasoning.
- Q&A session.

10:30 AM – 11:15 AM: Panel Discussion (Hybrid)

- Panel with experts in AI, HCI, cognitive and social science.
- Tentative panelists: Prof. Pattie Maes, known for her work on AI agents that reduce work and information overload; Prof. Lucas Graves, known for his work on journalistic methods for fact-checking and information evaluation; and Dr. Paolo Torroni.
- Moderator-led discussion on interdisciplinary approaches to AI-augmented reasoning.

11:15 AM – 12:00 PM: Paper Session 1 (Hybrid)

- Presentations of accepted papers (10 minutes each).
- Q&A session.

12:00 PM – 1:00 PM: Lunch Break

1:00 PM – 2:00 PM: Group Activity 1 (Hybrid)

- Breakout sessions focusing on specific topics discussed in the panel.
- Identify challenges, opportunities, and future directions.

2:00 PM – 2:15 PM: Coffee Break (Hybrid)

2:15 PM – 3:30 PM: Paper Session 2 (Hybrid)

- Presentations of accepted papers (15 minutes each).
- Q&A session.

3:30 PM – 4:30 PM: Group Activity 2 (Hybrid)

- Design exercise where participants create prototypes of AI-augmented reasoning systems.
- Presentations and feedback.

4:30 PM – 4:45 PM: Closing Remarks (Hybrid)

- Summary of key insights and outcomes.
- Discussion on future directions and potential collaborations.
- Closing remarks and next steps.

The aim is to create an environment that fosters in-depth discussions and collaborations between all participants, and to explore new ideas and approaches in AI-augmented reasoning. Given the multidisciplinary nature of this area, the workshop will provide a unique platform for researchers from various backgrounds to share, collaborate, and identify key open questions in AI-augmented reasoning. The workshop will be both in-person and virtual, allowing participation from an international audience. After the event, recordings and materials will be made available on the workshop's website for reference and further engagement.

7 Post-Workshop Plans

We plan to post photos and videos of the workshop on our website and social media during and after the workshop. We will encourage participants who have met through the workshop to submit articles together and comment on other participants' articles on the workshop's website. Our intention is that workshop participants will become co-authors to foster community and collaboration.

8 Call for Participation

AI-Augmented Reasoning systems are cognitive assistants that support human reasoning by providing AI-based feedback that can help users improve their critical reasoning skills. Made possible with new techniques like argumentation mining, fact-checking, crowdsourcing, and attention nudging, AI augmented reasoning systems can provide real-time feedback on logical reasoning, help users identify and avoid flawed arguments and misinformation, suggest counter-arguments, provide evidence-based explanations, and foster deeper reflection.

The goal of this workshop is to bring together researchers from AI, HCI, and cognitive science to discuss recent advances in AI-augmented reasoning and to identify open problems in this area. We invite submissions of short papers (2-4 pages) that describe original and unpublished works on AI-augmented reasoning. Submissions will be judged based on the following criteria: scientific soundness, novelty, technical depth, clarity of writing and presentation, and potential to impact and stimulate discussion. The workshop's website includes examples of past work in the area to help inspire and inform position papers. Papers should be a maximum of 4 pages, and should be submitted in the CHI Extended Abstracts format. The submission deadline is February 15th, 2025.

At least one author of each accepted position paper must attend the workshop and all participants must register for at least one day of the conference. We will host accepted papers on the workshop's website for participants and others to review prior to the meeting. Submission can be accessed through the website: http://aireasoning.media.mit.edu/.

9 Expected size of attendance

We expect 30-50 participants with at least 20 in-person registrations.

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